



Barker College

**2011
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 1

Staff Involved:

AM FRIDAY 12 AUGUST

- PJR* • GIC*
- MRB • GDH
- KJL • RMH
- GPF

105 copies

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages of your solutions
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- ALL necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value
- Marks may be deducted for careless or poorly arranged working

Total marks – 84
Attempt Questions 1–7
ALL questions are of equal value

Answer each question on a SEPARATE sheet of paper

Marks

Question 1 (12 marks) **[START A NEW PAGE]**

- (a) The point $P(x, y)$ divides the interval AB internally in the ratio $2 : 1$ **2**

If A is the point $(6, 1)$ and B is the point $(12, -8)$, find the coordinates of $P(x, y)$

- (b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\tan x}{3x} \right)$ **2**

- (c) Use the table of standard integrals to evaluate $\int_0^{\frac{\pi}{2}} \sec \frac{x}{2} \tan \frac{x}{2} dx$ **2**

- (d) Solve $\frac{x}{x-4} \leq 2$ **3**

- (e) Evaluate $\int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{dx}{\sqrt{1-4x^2}}$ **3**

Question 2 (12 marks) **[START A NEW PAGE]**

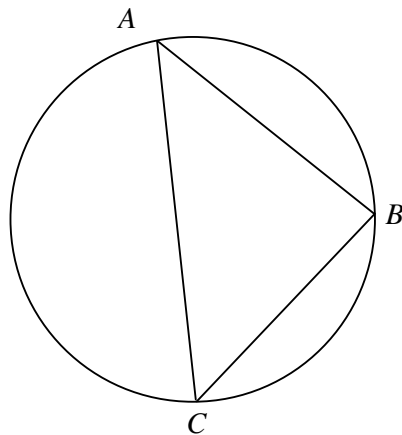
- (a) Find the acute angle between the curves $y = \log_e x$ and $y = 1 - x^2$ at the point P (1, 0) 3

Give your answer correct to the nearest minute.

- (b) The point $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$ with focus $S(0, a)$
- (i) Find M , the midpoint of the chord OP , where O is the origin 1
- (ii) Find the gradient of the chord OP 1
- (iii) Find the point A on the parabola where the tangent is parallel to the chord OP 2
- (iv) Show that A is equidistant from M and the x -axis 1

- (c) $\triangle ABC$ is inscribed in a circle as shown below.

The tangent at C meets AB produced at P and the bisector of $\angle ACB$ meets AB at Q



- (i) Copy and complete the diagram 1
- (ii) Prove that $PC = PQ$ 3

Question 3 (12 marks) **[START A NEW PAGE]**

- (a) Let $f(x) = \ln(\tan x)$, where $0 < x < \frac{\pi}{2}$ **3**

Show that $f'(x) = 2 \operatorname{cosec} 2x$

- (b) Use the substitution $x = 2 \sin \theta$ to evaluate $\int_0^1 \sqrt{4 - x^2} \, dx$ **3**

- (c) (i) State the domain and range of the function $f(x) = \cos^{-1} 2x$ **2**

- (ii) Draw a neat sketch of the function $f(x) = \cos^{-1} 2x$ **1**

Clearly label all essential features

- (iii) Find the equation of the tangent to the curve $f(x) = \cos^{-1} 2x$ at the point where the curve crosses the y-axis. **3**

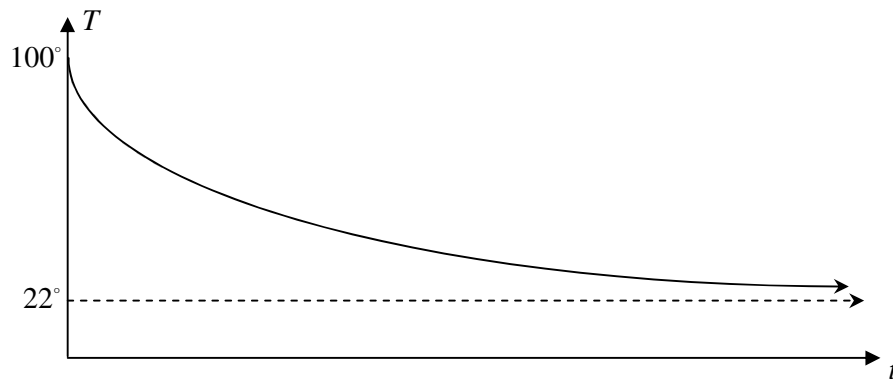
Question 4 (12 marks) **[START A NEW PAGE]**

(a) (i) Show that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ 1

(ii) Hence, or otherwise, evaluate $\int_0^{\pi/4} \sin 4x \cos 2x \, dx$ 3

(b) If $f(x + 2) = x^2 + 2$, find $f(x)$ 2

(c) The graph shown below represents the relationship between T , the temperature in $^{\circ}\text{C}$ of a cooling cup of coffee, and t , the time in minutes.



The rate of cooling of this coffee is given by $\frac{dT}{dt} = -k(T - A)$, where k and A are constants and $k > 0$.

(i) Show that $T = A + Be^{-kt}$ is a solution to the differential equation $\frac{dT}{dt} = -k(T - A)$, given that B is a constant. 1

(ii) By examining the graph when $t = 0$ and $t \rightarrow \infty$, find the values of A and B 2

(iii) If the temperature of the coffee is 50°C after 90 minutes, show that $k = -\frac{1}{90} \ln\left(\frac{14}{39}\right)$ 2

(iv) Hence, find the rate at which the coffee is cooling after 90 minutes. 1
Give your answer correct to two significant figures.

Question 5 (12 marks) **[START A NEW PAGE]**

- (a) Evaluate $\int_0^{\frac{\pi}{4}} \cos x \sin^2 x \, dx$ **2**
- (b) The volume of a sphere is increasing at the rate of 5 cm^3 per second. **3**
At what rate is the surface area increasing when the radius is 20 cm ?
- (c) A particle moves in such a way that its displacement x cm from an origin O at any time t seconds is given by the function $x = 4 + \sqrt{3} \cos 3t - \sin 3t$
- (i) Show that the particle is moving in simple harmonic motion. **2**
- (ii) Express $\sqrt{3} \cos 3t - \sin 3t$ in the form $R \cos(3t + \alpha)$, where α is acute and in radians. **2**
- (iii) Find the amplitude of the motion. **1**
- (iv) Find when the particle first passes through the centre of motion. **2**

Question 6 (12 marks) **[START A NEW PAGE]**

(a) Show by induction that $7^n + 2$ is divisible by 3, for all positive integers n **3**

(b) Given the function $f(x) = \frac{2x+1}{x-1}$

(i) Find any vertical and horizontal asymptotes **1**

(ii) State the domain of the inverse function $f^{-1}(x)$ **1**

(iii) Sketch the graph of the inverse function $f^{-1}(x)$ **2**
Clearly label all critical features of the inverse function $f^{-1}(x)$

(c) A particle is moving along the x -axis so that its acceleration after t seconds is given by

$$\ddot{x} = -e^{-\frac{x}{2}}$$

The particle starts at the origin with an initial velocity of 2 cm/sec

(i) If v is the velocity of the particle, find v^2 as a function of x **2**

(ii) Show that the displacement x as a function of time t is given by **3**

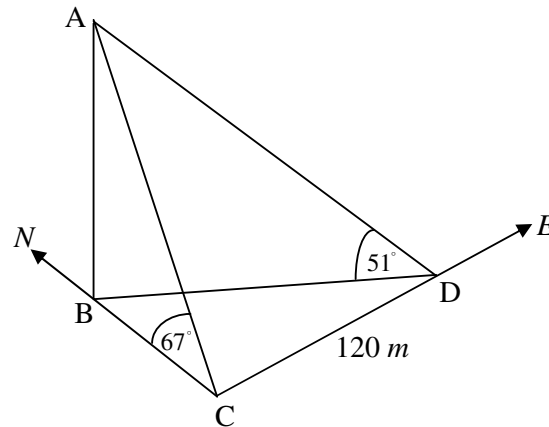
$$x = 4 \log_e \left(\frac{t+2}{2} \right)$$

Question 7 (12 marks) **[START A NEW PAGE]**

- (a) James is standing at the top A of a tower AB which is built on level ground.

From point C, due south of the base B of the tower, the angle of elevation of the top A of the tower is 67°

From point D, which is 120 m due east of point C, the angle of elevation of the top A of the tower is 51°



- (i) Calculate the height of the tower AB (to the nearest metre) **3**

- (ii) James projects a stone horizontally from the top of the tower with velocity $V\text{ m/s}$

If this stone lands at point D, find the value of V **3**

(Give your answer correct to one decimal place)

You may assume the equations of motion are

$$x = vt \cos \theta \text{ and } y = vt \sin \theta - 5t^2 \text{ (Do **NOT** prove this)}$$

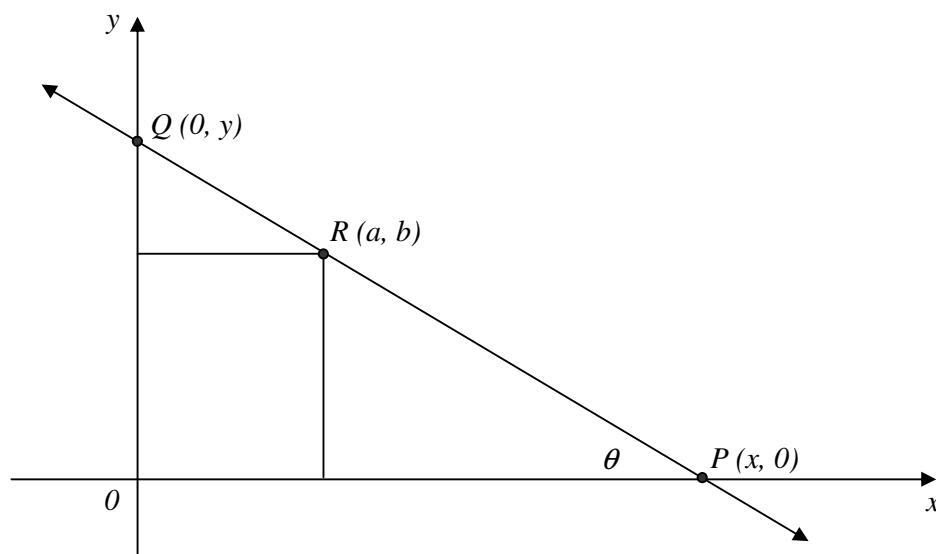
(Hint: Use point A as the origin)

Question 7 continues on page 9

Question 7 (continued)

- (b) The point $R(a, b)$ lies in the positive quadrant of the number plane.

A line through R meets the positive x and y axes at P and Q respectively and makes an angle θ with the x -axis.



- (i) Show that the length of PQ is equal to $\frac{a}{\cos \theta} + \frac{b}{\sin \theta}$ 2
- (ii) Hence, show that the minimum length of PQ is equal to $\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{3}{2}}$ 4

End of Question 7

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

AR 12 MATHEMATICS EXTENSION 1 TRIAL HSC

20 August 2011

Question 1

$$m:n = 2:1$$

$$= \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{2 \times 2 + 1 \times 6}{3}, \frac{2 \times -8 + 1 \times 1}{3} \right)$$

$$= \left(\frac{24+6}{3}, \frac{-16+1}{3} \right)$$

$$= (10, -5)$$

$$\frac{1}{3} \lim_{x \rightarrow 0} \frac{\tan x}{x} = \frac{1}{3} \times 1 = \frac{1}{3}$$

$$\left[2 \sec \frac{x}{2} \right]_0^{\frac{\pi}{2}}$$

$$2 \left(\sec \frac{\pi}{4} - \sec 0 \right)$$

$$2 \times (\sqrt{2} - 1)$$

$$= 2(\sqrt{2} - 1)$$

1st method:

multiplying both sides by the square of the denominator

$$(x-4)^2 \times \frac{x}{x-4} \leq 2(x-4)^2$$

$$(x-4)x \leq 2(x^2 - 8x + 16)$$

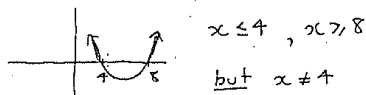
$$x^2 - 4x \leq 2x^2 - 16x + 32$$

$$0 \leq x^2 - 12x + 32$$

$$\therefore x^2 - 12x + 32 \geq 0$$

$$(x-4)(x-8) \geq 0$$

zeros are $x=4, x=8$



but $x \neq 4$
because it is a zero of the denominator

$$\therefore x < 4 \text{ or } x > 8$$

2nd Method: graphical

$$\text{for } \frac{x}{x-4} \leq 2$$

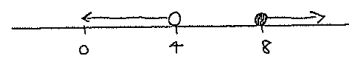
$$1. x \neq 4$$

$$2. \text{ solve } \frac{x}{x-4} = 2$$

$$x = 2x - 8$$

$$x = 8$$

$\therefore$ critical points are $x=4, 8$



check all 3 regions:

$$i) x < 4, x=0, \frac{0}{0-4} \leq 2$$

$$\text{true } \therefore x < 4$$

$$ii) 4 < x < 8, x=5, \frac{5}{5-4} \leq 2$$

not true

$$iii) x > 8, x=9, \frac{9}{9-4} \leq 2$$

$$\therefore x < 4 \text{ or } x > 8$$

$$e) \int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{dx}{\sqrt{1-(2x)^2}}$$

$$= \frac{1}{2} \int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{2 dx}{\sqrt{1-(2x)^2}}$$

$$= \frac{1}{2} \left[\sin^{-1} 2x \right]_{-\frac{1}{4}}^{\frac{1}{4}}$$

$$= \frac{1}{2} \left(\sin^{-1} \frac{1}{2} - \sin^{-1} \left(-\frac{1}{2} \right) \right)$$

$$= \frac{1}{2} \left(\sin^{-1} \frac{1}{2} + \sin^{-1} \frac{1}{2} \right)$$

$$= \frac{1}{2} \times 2 \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{6}$$

Question 2

$$a) \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$y_1 = \log_e x$$

$$y_2 = 1 - x^2$$

$$y'_1 = \frac{1}{x}$$

$$y'_2 = -2x$$

$$\text{at } x=1, m_1 = y'_1 = \frac{1}{1} = 1$$

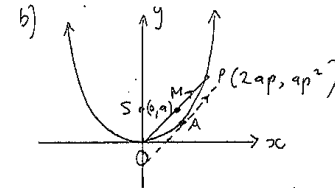
$$x=1, m_2 = y'_2 = -2 \times 1 = -2$$

$$\therefore \tan \theta = \left| \frac{1 - (-2)}{1 + (-2)} \right|$$

$$= \left| \frac{3}{-1} \right|$$

$$\tan \theta = 3$$

$$\theta = \tan^{-1} 3 = 71^\circ 34'$$



$$i) M = \left(\frac{0+2ap}{2}, \frac{0+ap^2}{2} \right)$$

$$M = \left(ap, \frac{ap^2}{2} \right)$$

$$ii) \text{ Gradient of } OP = \frac{ap^2 - 0}{2ap - 0}$$

$$\therefore m_{OP} = \frac{p}{2}$$

iii) A lies on the parabola

$$x^2 = 4ay \therefore y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a} = \frac{x}{2a}$$

now $y' = \frac{p}{2}$ since chords are parallel

$$\frac{x}{2a} = \frac{p}{2}$$

$$x = ap$$

$$\text{if } x = ap, y = \frac{(ap)^2}{4a} = \frac{ap^2}{4}$$

$$\therefore A \left(ap, \frac{ap^2}{4} \right)$$

iv) Distance of A from x-axis

= its y value

$$\therefore d_1 = \frac{ap^2}{4} \text{ units}$$

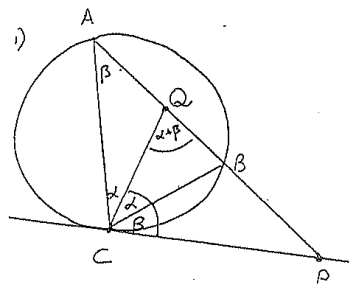
Distance from $A(ap, \frac{ap}{4})$

$M(ap, \frac{ap}{2})$ is

$$\sqrt{0 + \left(\frac{ap}{4}\right)^2}$$

$\frac{ap}{4}$ = distance of M from x -axis

A is equidistant from M and the x -axis.



To prove that $PC = PQ$
It is easiest to prove that $\angle QCP = \angle QAP$

$\therefore \angle QCA = \angle QCB = \alpha$ since CA is the bisector

Let $\angle BCP = \beta$

$\therefore \angle CAB = \beta$ (angle in the alternate segment equals angle between chord and tangent)

now $\angle QAP = \alpha + \beta$ (exterior angle of $\triangle AQC$)

$\angle QCP = \alpha + \beta$ (adjacent angles)

$\therefore \angle QAP = \angle QCP$

$\therefore \triangle PCA$ is isosceles (base angles are equal)

$\therefore PC = PQ$ (equal sides in isosceles $\triangle$)

Question 3

a) $f'(x) = \sec^2 x \leftarrow f'(x)$
LHS = $\frac{1}{\tan x} \times \sec^2 x$

$$= \frac{1}{\frac{\sin x}{\cos x}} \times \frac{1}{\cos^2 x}$$

$$= \frac{1}{\cos x} \times \frac{\cos x}{\sin x}$$

$$= \frac{1}{\cos x} \times \frac{1}{\sin x}$$

$$= \frac{2}{2 \cos x \sin x}$$

$$= \frac{2}{\sin 2x}$$

$$= 2 \operatorname{cosec} 2x$$

$$= \text{RHS}$$

b) $x = 2 \sin \theta$

$$dx = 2 \cos \theta d\theta$$

when $x=0$, $\theta=0$

$$x=1, \sin \theta = \frac{1}{2}, \theta = \frac{\pi}{6}$$

$$\therefore \int_0^{\frac{\pi}{6}} \sqrt{4 - (2 \sin \theta)^2} \times 2 \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{4(1 - \sin^2 \theta)} \times 2 \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{4 \cos^2 \theta} \times 2 \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} 2 \cos \theta \times 2 \cos \theta d\theta$$

$$= 4 \int_0^{\frac{\pi}{6}} \cos^2 \theta d\theta$$

$$\text{now } \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\therefore = 4 \int_0^{\frac{\pi}{6}} \frac{1 + \cos 2\theta}{2} d\theta$$

$$= 2 \int_0^{\frac{\pi}{6}} 1 + \cos 2\theta d\theta$$

$$= 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{6}}$$

$$= 2 \left(\frac{\pi}{6} + \frac{\sin \frac{\pi}{3}}{2} - 0 - 0 \right)$$

$$= 2 \left(\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right)$$

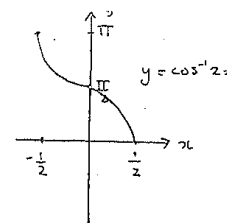
$$= \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

c) i) $f(x) = \cos^2 2x$

Domain: $-1 \leq 2x \leq 1$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

Range: $0 \leq y \leq 1$



ii) Gradient of tangent

$$f'(x) = \frac{-2}{\sqrt{1 - (2x)^2}}$$

$$= \frac{-2}{\sqrt{1 - 4x^2}}$$

at $(0, \frac{\pi}{2})$ $f'(0) = \frac{-2}{1} = -2 = m$

$\therefore$ Eqn of tangent

$$y - \frac{\pi}{2} = -2(x - 0)$$

$$y - \frac{\pi}{2} = -2x$$

$$2x + y - \frac{\pi}{2} = 0$$

$$(\text{or } y = -2x + \frac{\pi}{2})$$

Question 4

a) i) LHS = $\sin A \cos B + \cos A \sin B$
 $+ \sin A \cos B - \cos A \sin B$

$$= 2 \sin A \cos B$$

$$= \text{RHS}$$

ii) $\frac{1}{2} \int_0^{\frac{\pi}{4}} 2 \sin 4x \cos 2x dx$

$$A = 4x \quad B = 2x$$

$$\therefore A+B = 6x \quad A-B = 2x$$

$$\frac{1}{2} \int_0^{\frac{\pi}{4}} \sin 6x + \sin 2x \, dx$$

$$\frac{1}{2} \left[\frac{-\cos 6x}{6} - \frac{\cos 2x}{2} \right]_0^{\frac{\pi}{4}}$$

$$= -\frac{1}{2} \left(\frac{\cos \frac{3\pi}{2}}{6} + \frac{\cos \frac{\pi}{2}}{2} - \left(\frac{\cos 0}{6} + \frac{\cos 0}{2} \right) \right)$$

$$= -\frac{1}{2} \left(0 + 0 - \left(\frac{1}{6} + \frac{1}{2} \right) \right)$$

$$= -\frac{1}{2} \times -\frac{2}{3}$$

$$= \frac{1}{3}$$

1st method:

$$f(x+2) = (x+2)^2 - 4x - 2$$

↑
rewrite RHS in terms of $(x+2)^2$

$$f(x+2) = (x+2)^2 - 4(x+2) + 6 \quad \therefore x+2 = (x+2)^2 - 4(x+2) + 6$$

$$\therefore f(x) = x^2 - 4x + 6$$

(replace $(x+2)$ with x)

2nd method:

$$f(x) = f((x-2)+2)$$

$$= (x-2)^2 + 2$$

$$= x^2 - 4x + 4 + 2$$

$$= x^2 - 4x + 6$$

3rd method:

$$\text{Equate } x^2 + 2 \equiv A(x+2)^2 + B(x+2) + C$$

$$\text{sub } x = -2$$

$$6 \equiv 0 + 0 + C$$

$$C = 6$$

$$\text{sub } x = 0, \quad 2 = 4A + 2B + C$$

$$2 = 4A + 2B + 6$$

$$2A + B = -2$$

$$\text{sub } x = -1, \quad 3 = A + B + C$$

$$A + B = -3$$

$\therefore$ solve simultaneously

$$2A + B = -2 \quad \text{①}$$

$$A + B = -3 \quad \text{②}$$

$$\text{①} - \text{②} \quad A = 1$$

$$\therefore B = -4$$

$$\therefore f(x+2) = (x+2)^2 - 4(x+2) + 6$$

$$f(x) = x^2 - 4x + 6$$

c) i) Method 1

$$T = A + Be^{-kt}$$

$$\frac{dT}{dt} = -kBe^{-kt}$$

$$= -k(A + Be^{-kt} - A)$$

$$= -k(T - A)$$

$$\therefore \frac{dT}{dt} = -k(T - A)$$

Method 2

$$T = A + Be^{-kt}$$

$$\text{show that } \frac{dT}{dt} = -k(T - A)$$

$$\text{LHS} = \frac{dT}{dt} = -kBe^{-kt}$$

$$\text{RHS} = -k(A + Be^{-kt} - A)$$

$$= -k(Be^{-kt})$$

$$= -kBe^{-kt}$$

$$= \text{LHS}$$

ii) from graph $t=0, T=100^\circ$

$$\text{sub } t=0, \quad 100 = A + Be^0$$

$$A + B = 100$$

$$t \rightarrow \infty, T \rightarrow 22^\circ$$

$$\text{as } t \rightarrow \infty \quad Be^{-kt} \rightarrow 0$$

$$22 = A + 0$$

$$\therefore A = 22$$

$$\text{and } B = 78$$

$$\therefore T = 22 + 78e^{-kt}$$

iii) $T = 50, t = 90$

$$50 = 22 + 78e^{-90k}$$

$$28 = 78e^{-90k}$$

$$e^{-90k} = \frac{14}{39}$$

$$-90k = \log_e \left(\frac{14}{39} \right)$$

$$k = -\frac{1}{90} \log_e \left(\frac{14}{39} \right)$$

iv) Rate is

$$\frac{dT}{dt} = -k(T - A)$$

$$\frac{dT}{dt} = \frac{1}{90} \log_e \left(\frac{14}{39} \right) (50 - 22)$$

$$= \frac{1}{90} \log_e \left(\frac{14}{39} \right) \times 28$$

$$= -0.318 T \dots$$

$$= -0.32 \text{ } ^\circ\text{C/min}$$

(to 2 sig figs)

Question 5

$$a) \left[\frac{5\omega^3 x}{3} \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{3} (5\omega^3 \frac{\pi}{4} - 5\omega^3 0)$$

$$= \frac{1}{3} \times \left(\frac{1}{\sqrt{2}} \right)^3$$

$$= \frac{1}{3} \times \frac{1}{2\sqrt{2}}$$

$$= \frac{1}{6\sqrt{2}} \quad (\text{or } \frac{\sqrt{2}}{12})$$

$$b) \frac{dV}{dt} = 5 \text{ cm}^3/\text{s}$$

$$\text{now } \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\therefore 5 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{5}{4\pi r^2}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\frac{dA}{dt} = 8\pi r \times \frac{dr}{dt}$$

$$= 8\pi \times \frac{5}{4\pi r}$$

$$= \frac{10}{r}$$

when $r = 20$ cm

$$\frac{dA}{dt} = \frac{10}{20} = \frac{1}{2}$$

∴ rate at which surface area is increasing is $\frac{1}{2} \text{ cm}^2/\text{s}$

c) i) $x = 4 + \sqrt{3} \cos 3t - \sin 3t$

$$\dot{x} = -3\sqrt{3} \sin 3t - 3 \cos 3t$$

$$\ddot{x} = -9\sqrt{3} \cos 3t + 9 \sin 3t$$

$$\ddot{x} = -9(\sqrt{3} \cos 3t - \sin 3t + 4) + 36$$

$$\ddot{x} = -9x + 36$$

$$= -9(x-4)$$

which is SHM, $n=3$, centre is 4 cm

$$\sqrt{3} \cos 3t - \sin 3t = R \cos 3t \cos \alpha - R \sin 3t \sin \alpha$$

Equating both sides:

$$\left. \begin{aligned} R \cos \alpha &= \sqrt{3} \\ R \sin \alpha &= 1 \end{aligned} \right\} \begin{aligned} \text{①} \\ \text{②} \end{aligned}$$

② ÷ ① $\tan \alpha = \frac{1}{\sqrt{3}}$

$$\alpha = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$R = \sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 + 1^2} = \sqrt{4} = 2$$

$$\therefore 2 \cos\left(3t + \frac{\pi}{6}\right)$$

iii) so, $x = 4 + 2 \cos\left(3t + \frac{\pi}{6}\right)$

$$\text{since } -1 \leq \cos\left(3t + \frac{\pi}{6}\right) \leq 1$$

then x can be between $(4+2)$ cm and $(4-2)$ cm

$$\text{i.e. } 2 \leq x \leq 6$$

the centre is $x = 4$

∴ amplitude is 2 cm

iv) solve:

$$4 + 2 \cos\left(3t + \frac{\pi}{6}\right) = 4$$

$$2 \cos\left(3t + \frac{\pi}{6}\right) = 0$$

$$\cos\left(3t + \frac{\pi}{6}\right) = 0$$

$$3t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

1st time

$$\therefore 3t + \frac{\pi}{6} = \frac{\pi}{2}$$

$$3t = \frac{\pi}{3}$$

$$t = \frac{\pi}{9}$$

∴ particle first passes

through $x=4$ after $t = \frac{\pi}{9}$ sec

Question 6

a) show that $7^n + 2 = 3N$

where N and n are integers both > 1

Prove true for $n=1$

$$\text{LHS} = 7^1 + 2 = 9 = 3 \times 3$$

$$= \text{RHS for } N=3$$

∴ statement true for $n=1$

Assume true for $n=k$ where

k is an integer > 1

$$\text{i.e. } 7^k + 2 = 3N$$

now prove true for $n=k+1$

$$7^{k+1} + 2 = 3M \text{ where } M \text{ is a positive integer}$$

$$\text{LHS} = 7^{k+1} + 2$$

$$= 7 \times 7^k + 2$$

$$= 7 \times (3N - 2) + 2 \quad (\text{from assumption})$$

$$= 21N - 14 + 2$$

$$= 21N - 12$$

$$= 3(7N - 4)$$

$$= 3M \text{ where } M = 7N - 4$$

$$= \text{RHS}$$

∴ statement is true for $n=k+1$

∴ statement is true for $n=1$,

$n=k$ and $n=k+1$

∴ it is true for all positive integers n

b) i) $f(x) = \frac{2x+1}{x-1}$

vertical asymptote $x=1$

(as $x-1 \neq 0$)

horizontal asymptote

$$\lim_{x \rightarrow \infty} \frac{2x+1}{x-1} = \lim_{x \rightarrow \infty} \frac{\frac{2x}{x} + \frac{1}{x}}{\frac{x}{x} - \frac{1}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{2 + \frac{1}{x}}{1 - \frac{1}{x}}$$

$$= \frac{2+0}{1-0}$$

$$= 2$$

∴ $y=2$ is horiz. asymptote

ii) Domain of $f^{-1}(x)$ is the same as range of $f(x)$
∴ $x \neq 2$

iii) Sketch $y = f^{-1}(x)$

$$D: x \neq 2$$

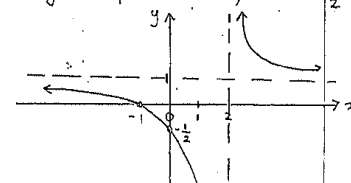
$$R: y \neq 1$$

for $y=f(x)$ the y -intercept is -1

∴ x intercept of $f^{-1}(x)$ is -1

for $y=f(x)$ the x -intercept is $-\frac{1}{2}$

∴ y intercept of $f^{-1}(x)$ is $-\frac{1}{2}$



$$\begin{aligned}
 i) \quad \frac{1}{2} V^2 &= \int -e^{-\frac{x}{2}} dx \\
 \frac{1}{2} V^2 &= -\frac{e^{-\frac{x}{2}}}{-\frac{1}{2}} + C \\
 \frac{1}{2} V^2 &= 2e^{-\frac{x}{2}} + C
 \end{aligned}$$

$$= 0, v = 2$$

$$2 = 2 + C$$

$$\therefore C = 0$$

$$\therefore V^2 = 4e^{-\frac{x}{2}}$$

$$V = \pm \sqrt{4e^{-\frac{x}{2}}}$$

$$V = \pm 2e^{-\frac{x}{4}}$$

$$\text{at } x=0, V=2 \therefore \text{take}$$

positive V

$$V = 2e^{-\frac{x}{4}}$$

$$\frac{dx}{dt} = 2e^{-\frac{x}{4}} = \frac{2}{e^{\frac{x}{4}}}$$

$$\therefore \int \frac{e^{\frac{x}{4}}}{2} dx = \int dt$$

$$\frac{1}{2} \left(\frac{e^{\frac{x}{4}}}{\frac{1}{4}} \right) = t + K$$

$$\frac{1}{2} \times 4e^{\frac{x}{4}} = t + K$$

$$2e^{\frac{x}{4}} = t + K$$

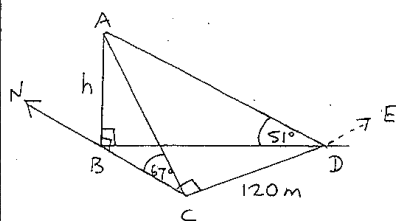
$$t=0, x=0 \quad 2 = 0 + K$$

$$\therefore K = 2$$

$$2e^{\frac{x}{4}} = t + 2$$

$$\begin{aligned}
 e^{\frac{x}{4}} &= \frac{t+2}{2} \\
 \frac{x}{4} &= \log_e \left(\frac{t+2}{2} \right) \\
 \therefore x &= 4 \log_e \left(\frac{t+2}{2} \right)
 \end{aligned}$$

Question 7



$$i) \tan 67^\circ = \frac{h}{BC} \quad \tan 51^\circ = \frac{h}{BD}$$

$$BC = \frac{h}{\tan 67^\circ} \quad BD = \frac{h}{\tan 51^\circ}$$

$$BC = h \tan 23^\circ \quad BD = h \tan 39^\circ$$

In the base $\triangle BDC$:

use Pythagoras' Theorem since

$$\angle BCD = 90^\circ$$

$$BD^2 = BC^2 + 120^2$$

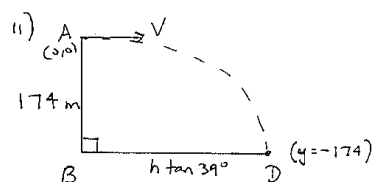
$$(h \tan 39^\circ)^2 = (h \tan 23^\circ)^2 + 120^2$$

$$h^2 (\tan^2 39^\circ - \tan^2 23^\circ) = 120^2$$

$$h^2 = \frac{120^2}{\tan^2 39^\circ - \tan^2 23^\circ}$$

$$h = \frac{120}{\sqrt{\tan^2 39^\circ - \tan^2 23^\circ}}$$

$$\therefore h = 174 \text{ m (nearest m)}$$



horizontal projection $\therefore \theta = 0$

$$x = Vt \cos \theta, \quad y = Vt \sin \theta - 5t^2$$

$$\theta = 0$$

$$x = Vt, \quad y = -5t^2$$

$$\text{At point D, } y = -174$$

$$-174 = -5t^2$$

$$t^2 = \frac{174}{5}$$

$$t = \sqrt{\frac{174}{5}} = 5.9 \text{ sec}$$

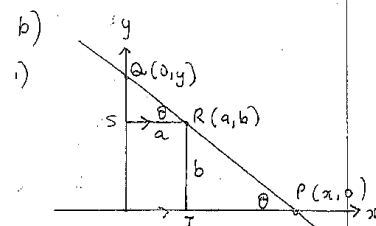
then $x = Vt$

$$h \tan 39^\circ = V \times 5.9$$

$$V = \frac{174 \times \tan 39^\circ}{5.9}$$

$$= 23.8817 \dots$$

$$\therefore V = 23.9 \text{ m/s}$$



$$i) \text{ length } PQ = QR + RP$$

$$\text{In } \triangle RTP: \sin \theta = \frac{b}{RP}$$

$$\therefore RP = \frac{b}{\sin \theta}$$

$$\text{In } \triangle QRS: \cos \theta = \frac{a}{RQ}$$

$$RQ = \frac{a}{\cos \theta}$$

$$\therefore PQ = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

$$ii) \text{ let } L = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

$$\text{ie } L = a \sec \theta + b \csc \theta$$

to minimise length, solve

$$\frac{dL}{d\theta} = 0 \text{ for } \theta$$

$$L' = a \sec \theta \tan \theta + -b \csc \theta \cot \theta$$

$$= a \times \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta} - b \times \frac{1}{\sin \theta} \times \frac{\cos \theta}{\sin \theta}$$

$$= \frac{a \sin \theta}{\cos^2 \theta} - \frac{b \cos \theta}{\sin^2 \theta} = 0 \text{ for min}$$

$$\therefore \frac{a \sin \theta}{\cos^2 \theta} = \frac{b \cos \theta}{\sin^2 \theta}$$

$$a \sin^3 \theta = b \cos^3 \theta$$

$$\therefore \tan^3 \theta = \frac{b}{a}$$

$$\tan \theta = \sqrt[3]{\frac{b}{a}} = \frac{b^{\frac{1}{3}}}{a^{\frac{1}{3}}}$$

$$\theta = \tan^{-1}\left(\sqrt[3]{\frac{b}{a}}\right)$$

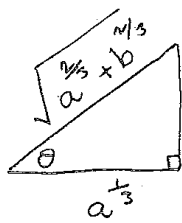
$$\therefore \theta = \tan^{-1}\left(\sqrt[3]{\frac{b}{a}}\right) \text{ gives the}$$

minimum length of PQ

(* it is a min. length since there is no max. value for

PQ, since as $\theta \rightarrow 0$, length PQ $\rightarrow \infty$)

$$\text{now, since } \tan \theta = \frac{b^{\frac{1}{3}}}{a^{\frac{1}{3}}}$$



use Pythagoras

$$\therefore \sin \theta = \frac{b^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}$$

$$\cos \theta = \frac{a^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}$$

$\therefore$ Minimum length of PQ:

$$PQ = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

$$= \frac{a}{\frac{a^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}} + \frac{b}{\frac{b^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}}$$

$$= \frac{a^{\frac{2}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}} + \frac{b^{\frac{2}{3}}}{\sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}$$

$$PQ = a^{\frac{2}{3}} \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}} + b^{\frac{2}{3}} \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}$$

$$= \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}} \left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)$$

$$= \left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{1}{2}} \left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)$$

$$= \left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{3}{2}}$$

!! when!

$\therefore$ min length of PQ is

equal to $\left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{3}{2}}$